

SuperKnob配置指南

在克隆远程仓库之后，platformio中导入TFT_eSPI库和lvgl库，需要对这两个库中的其中两个文件进行配置才能正常编译并使显示屏正常工作

参考视频https://www.bilibili.com/video/BV1jvh7z3E9W/?vd_source=bc76eeb177b6e51d565be82f8d14a9e5

视频中的屏幕和实际使用的屏幕驱动和尺寸不同，实际使用的屏幕为ST7789驱动，240*320像素

需要修改的配置文件

1. superknob_demo/.pio/libdeps/lolin32_lite/TFT_eSPI/User_Setup.h 编辑这个文件使之适配显示屏（ST7789驱动，240*320像素）

User_Setup.h

```
1  //                                USER DEFINED SETTINGS
2  //  Set driver type, fonts to be loaded, pins used and SPI control method etc
3  //
4  //  See the User_Setup_Select.h file if you wish to be able to define multiple
5  //  setups and then easily select which setup file is used by the compiler.
6  //
7  //  If this file is edited correctly then all the library example sketches
8  //  should run without the need to make any more changes for a particular hardware
9  //  setup!
10 //  Note that some sketches are designed for a particular TFT pixel
11 //  width/height
12
13 // User defined information reported by "Read_User_Setup" test & diagnostics
14 // example
15 #define USER_SETUP_INFO "User_Setup"
16
17 // Define to disable all #warnings in library (can be put in
18 // User_Setup_Select.h)
19 // #define DISABLE_ALL_LIBRARY_WARNINGS
20
21 //
22 #####
23 ###
24
25 //
26 // Section 1. Call up the right driver file and any options for it
27 //
```

```

21  //
    #####
    ###
22
23  // Define STM32 to invoke optimised processor support (only for STM32)
24  //#define STM32
25
26  // Defining the STM32 board allows the library to optimise the performance
27  // for UNO compatible "MCUfriend" style shields
28  //#define NUCLEO_64_TFT
29  //#define NUCLEO_144_TFT
30
31  // STM32 8 bit parallel only:
32  // If STM32 Port A or B pins 0-7 are used for 8 bit parallel data bus bits 0-7
33  // then this will improve rendering performance by a factor of ~8x
34  //#define STM_PORTA_DATA_BUS
35  //#define STM_PORTB_DATA_BUS
36
37  // Tell the library to use parallel mode (otherwise SPI is assumed)
38  //#define TFT_PARALLEL_8_BIT
39  //#define TFT_PARALLEL_16_BIT // **** 16 bit parallel ONLY for RP2040
    processor ****
40
41  // Display type - only define if RPi display
42  //#define RPI_DISPLAY_TYPE // 20MHz maximum SPI
43
44  // Only define one driver, the other ones must be commented out
45  // #define ILI9341_DRIVER // Generic driver for common displays
46  // #define ILI9341_2_DRIVER // Alternative ILI9341 driver, see
    https://github.com/Bodmer/TFT_eSPI/issues/1172
47  // #define ST7735_DRIVER // Define additional parameters below for this
    display
48  // #define ILI9163_DRIVER // Define additional parameters below for this
    display
49  // #define S6D02A1_DRIVER
50  // #define RPI_ILI9486_DRIVER // 20MHz maximum SPI
51  // #define HX8357D_DRIVER
52  // #define ILI9481_DRIVER
53  // #define ILI9486_DRIVER
54  // #define ILI9488_DRIVER // WARNING: Do not connect ILI9488 display SDO to
    MISO if other devices share the SPI bus (TFT SDO does NOT tristate when CS is
    high)
55  #define ST7789_DRIVER // Full configuration option, define additional
    parameters below for this display
56  // #define ST7789_2_DRIVER // Minimal configuration option, define
    additional parameters below for this display
57  // #define R61581_DRIVER

```

```
58  //#define RM68140_DRIVER
59  //#define ST7796_DRIVER
60  //#define SSD1351_DRIVER
61  //#define SSD1963_480_DRIVER
62  //#define SSD1963_800_DRIVER
63  //#define SSD1963_800ALT_DRIVER
64  //#define ILI9225_DRIVER
65  //#define GC9A01_DRIVER
66
67  // Some displays support SPI reads via the MISO pin, other displays have a
single
68  // bi-directional SDA pin and the library will try to read this via the MOSI
line.
69  // To use the SDA line for reading data from the TFT uncomment the following
line:
70
71  // #define TFT_SDA_READ      // This option is for ESP32 ONLY, tested with
ST7789 and GC9A01 display only
72
73  // For ST7735, ST7789 and ILI9341 ONLY, define the colour order IF the blue
and red are swapped on your display
74  // Try ONE option at a time to find the correct colour order for your display
75
76  #define TFT_RGB_ORDER TFT_RGB  // Colour order Red-Green-Blue
77  // #define TFT_RGB_ORDER TFT_BGR  // Colour order Blue-Green-Red
78
79  // For M5Stack ESP32 module with integrated ILI9341 display ONLY, remove // in
line below
80
81  // #define M5STACK
82
83  // For ST7789, ST7735, ILI9163 and GC9A01 ONLY, define the pixel width and
height in portrait orientation
84  // #define TFT_WIDTH  80
85  // #define TFT_WIDTH  128
86  // #define TFT_WIDTH  172 // ST7789 172 x 320
87  #define TFT_WIDTH  240 // ST7789 240 x 240 and 240 x 320
88  // #define TFT_HEIGHT 160
89  // #define TFT_HEIGHT 128
90  // #define TFT_HEIGHT 240 // ST7789 240 x 240
91  #define TFT_HEIGHT 320 // ST7789 240 x 320
92  // #define TFT_HEIGHT 240 // GC9A01 240 x 240
93
94  // For ST7735 ONLY, define the type of display, originally this was based on
the
95  // colour of the tab on the screen protector film but this is not always true,
so try
```

```

96 // out the different options below if the screen does not display graphics
    correctly,
97 // e.g. colours wrong, mirror images, or stray pixels at the edges.
98 // Comment out ALL BUT ONE of these options for a ST7735 display driver, save
    this
99 // this User_Setup file, then rebuild and upload the sketch to the board again:
100
101 // #define ST7735_INITB
102 // #define ST7735_GREENTAB
103 // #define ST7735_GREENTAB2
104 // #define ST7735_GREENTAB3
105 // #define ST7735_GREENTAB128    // For 128 x 128 display
106 // #define ST7735_GREENTAB160x80 // For 160 x 80 display (BGR, inverted, 26
    offset)
107 // #define ST7735_ROBOTLCD        // For some RobotLCD arduino shields
    (128x160, BGR, https://docs.arduino.cc/retired/getting-started-guides/TFT)
108 // #define ST7735_REDTAB
109 // #define ST7735_BLACKTAB
110 // #define ST7735_REDTAB160x80    // For 160 x 80 display with 24 pixel offset
111
112 // If colours are inverted (white shows as black) then uncomment one of the
    next
113 // 2 lines try both options, one of the options should correct the inversion.
114
115 // #define TFT_INVERSION_ON
116 // #define TFT_INVERSION_OFF
117
118 //
    #####
    ###
119 //
120 // Section 2. Define the pins that are used to interface with the display here
121 //
122 //
    #####
    ###
123
124 // If a backlight control signal is available then define the TFT_BL pin in
    Section 2
125 // below. The backlight will be turned ON when tft.begin() is called, but the
    library
126 // needs to know if the LEDs are ON with the pin HIGH or LOW. If the LEDs are
    to be
127 // driven with a PWM signal or turned OFF/ON then this must be handled by the
    user
128 // sketch. e.g. with digitalWrite(TFT_BL, LOW);
129

```

```

130 // #define TFT_BL 32 // LED back-light control pin
131 // #define TFT_BACKLIGHT_ON HIGH // Level to turn ON back-light (HIGH or LOW)
132
133 // We must use hardware SPI, a minimum of 3 GPIO pins is needed.
134 // Typical setup for ESP8266 NodeMCU ESP-12 is :
135 //
136 // Display SDO/MISO to NodeMCU pin D6 (or leave disconnected if not reading
    TFT)
137 // Display LED to NodeMCU pin VIN (or 5V, see below)
138 // Display SCK to NodeMCU pin D5
139 // Display SDI/MOSI to NodeMCU pin D7
140 // Display DC (RS/A0) to NodeMCU pin D3
141 // Display RESET to NodeMCU pin D4 (or RST, see below)
142 // Display CS to NodeMCU pin D8 (or GND, see below)
143 // Display GND to NodeMCU pin GND (0V)
144 // Display VCC to NodeMCU 5V or 3.3V
145 //
146 // The TFT RESET pin can be connected to the NodeMCU RST pin or 3.3V to free
    up a control pin
147 //
148 // The DC (Data Command) pin may be labelled A0 or RS (Register Select)
149 //
150 // With some displays such as the ILI9341 the TFT CS pin can be connected to
    GND if no more
151 // SPI devices (e.g. an SD Card) are connected, in this case comment out the
    #define TFT_CS
152 // line below so it is NOT defined. Other displays such as the ST7735 require
    the TFT CS pin
153 // to be toggled during setup, so in these cases the TFT_CS line must be
    defined and connected.
154 //
155 // The NodeMCU D0 pin can be used for RST
156 //
157 //
158 // Note: only some versions of the NodeMCU provide the USB 5V on the VIN pin
159 // If 5V is not available at a pin you can use 3.3V but backlight brightness
160 // will be lower.
161
162 // ##### EDIT THE PIN NUMBERS IN THE LINES FOLLOWING TO SUIT YOUR ESP8266
    SETUP #####
163
164 // For NodeMCU - use pin numbers in the form PIN_Dx where Dx is the NodeMCU
    pin designation
165 // #define TFT_CS PIN_D8 // Chip select control pin D8
166 // #define TFT_DC PIN_D3 // Data Command control pin
167 // #define TFT_RST PIN_D4 // Reset pin (could connect to NodeMCU RST, see
    next line)

```

```
168  //#define TFT_RST -1    // Set TFT_RST to -1 if the display RESET is  
connected to NodeMCU RST or 3.3V  
169  
170  //#define TFT_BL PIN_D1  // LED back-light (only for ST7789 with backlight  
control pin)  
171  
172  //#define TOUCH_CS PIN_D2    // Chip select pin (T_CS) of touch screen  
173  
174  //#define TFT_WR PIN_D2    // Write strobe for modified Raspberry Pi TFT  
only  
175  
176  // #####  FOR ESP8266 OVERLAP MODE EDIT THE PIN NUMBERS IN THE FOLLOWING  
LINES  #####  
177  
178  // Overlap mode shares the ESP8266 FLASH SPI bus with the TFT so has a  
performance impact  
179  // but saves pins for other functions. It is best not to connect MISO as some  
displays  
180  // do not tristate that line when chip select is high!  
181  // Note: Only one SPI device can share the FLASH SPI lines, so a SPI touch  
controller  
182  // cannot be connected as well to the same SPI signals.  
183  // On NodeMCU 1.0 SD0=MISO, SD1=MOSI, CLK=SCLK to connect to TFT in overlap  
mode  
184  // On NodeMCU V3  S0 =MISO, S1 =MOSI, S2 =SCLK  
185  // In ESP8266 overlap mode the following must be defined  
186  
187  //#define TFT_SPI_OVERLAP  
188  
189  // In ESP8266 overlap mode the TFT chip select MUST connect to pin D3  
190  //#define TFT_CS  PIN_D3  
191  //#define TFT_DC  PIN_D5  // Data Command control pin  
192  //#define TFT_RST  PIN_D4  // Reset pin (could connect to NodeMCU RST, see  
next line)  
193  //#define TFT_RST -1  // Set TFT_RST to -1 if the display RESET is connected  
to NodeMCU RST or 3.3V  
194  
195  // ##### EDIT THE PIN NUMBERS IN THE LINES FOLLOWING TO SUIT YOUR ESP32 SETUP  
#####  
196  
197  // For ESP32 Dev board (only tested with ILI9341 display)  
198  // The hardware SPI can be mapped to any pins  
199  
200  //#define TFT_MISO 19  
201  //#define TFT_MOSI 23  
202  //#define TFT_SCLK 18  
203  //#define TFT_CS  15  // Chip select control pin
```

```

204 // #define TFT_DC    2  // Data Command control pin
205 // #define TFT_RST   4  // Reset pin (could connect to RST pin)
206 // #define TFT_RST  -1  // Set TFT_RST to -1 if display RESET is connected to
    ESP32 board RST
207
208 // For ESP32 Dev board (only tested with GC9A01 display)
209 // The hardware SPI can be mapped to any pins
210
211 #define TFT_MOSI 19 // In some display driver board, it might be written as
    "SDA" and so on.
212 #define TFT_SCLK 18
213 #define TFT_CS   16  // Chip select control pin
214 #define TFT_DC   15  // Data Command control pin
215 #define TFT_RST   13  // Reset pin (could connect to Arduino RESET pin)
216 #define TFT_BL   17  // LED back-light
217
218 // #define TOUCH_CS 21      // Chip select pin (T_CS) of touch screen
219
220 // #define TFT_WR 22      // Write strobe for modified Raspberry Pi TFT only
221
222 // For the M5Stack module use these #define lines
223 // #define TFT_MISO 19
224 // #define TFT_MOSI 23
225 // #define TFT_SCLK 18
226 // #define TFT_CS   14  // Chip select control pin
227 // #define TFT_DC   27  // Data Command control pin
228 // #define TFT_RST   33  // Reset pin (could connect to Arduino RESET pin)
229 // #define TFT_BL   32  // LED back-light (required for M5Stack)
230
231 // #####          EDIT THE PINs BELOW TO SUIT YOUR ESP32 PARALLEL TFT SETUP
    #####
232
233 // The library supports 8 bit parallel TFTs with the ESP32, the pin
234 // selection below is compatible with ESP32 boards in UNO format.
235 // Wemos D32 boards need to be modified, see diagram in Tools folder.
236 // Only ILI9481 and ILI9341 based displays have been tested!
237
238 // Parallel bus is only supported for the STM32 and ESP32
239 // Example below is for ESP32 Parallel interface with UNO displays
240
241 // Tell the library to use 8 bit parallel mode (otherwise SPI is assumed)
242 // #define TFT_PARALLEL_8_BIT
243
244 // The ESP32 and TFT the pins used for testing are:
245 // #define TFT_CS   33  // Chip select control pin (library pulls permanently
    low

```

```

246 // #define TFT_DC    15  // Data Command control pin - must use a pin in the
    range 0-31
247 // #define TFT_RST   32  // Reset pin, toggles on startup
248
249 // #define TFT_WR     4  // Write strobe control pin - must use a pin in the
    range 0-31
250 // #define TFT_RD     2  // Read strobe control pin
251
252 // #define TFT_D0     12  // Must use pins in the range 0-31 for the data bus
253 // #define TFT_D1     13  // so a single register write sets/clears all bits.
254 // #define TFT_D2     26  // Pins can be randomly assigned, this does not affect
255 // #define TFT_D3     25  // TFT screen update performance.
256 // #define TFT_D4     17
257 // #define TFT_D5     16
258 // #define TFT_D6     27
259 // #define TFT_D7     14
260
261 // #####          EDIT THE PINs BELOW TO SUIT YOUR STM32 SPI TFT SETUP
    #####
262
263 // The TFT can be connected to SPI port 1 or 2
264 // #define TFT_SPI_PORT 1 // SPI port 1 maximum clock rate is 55MHz
265 // #define TFT_MOSI  PA7
266 // #define TFT_MISO  PA6
267 // #define TFT_SCLK  PA5
268
269 // #define TFT_SPI_PORT 2 // SPI port 2 maximum clock rate is 27MHz
270 // #define TFT_MOSI  PB15
271 // #define TFT_MISO  PB14
272 // #define TFT_SCLK  PB13
273
274 // Can use Arduino pin references, arbitrary allocation, TFT_eSPI controls
    chip select
275 // #define TFT_CS     D5  // Chip select control pin to TFT CS
276 // #define TFT_DC     D6  // Data Command control pin to TFT DC (may be labelled
    RS = Register Select)
277 // #define TFT_RST    D7  // Reset pin to TFT RST (or RESET)
278 // OR alternatively, we can use STM32 port reference names PXnn
279 // #define TFT_CS     PE11 // Nucleo-F767ZI equivalent of D5
280 // #define TFT_DC     PE9  // Nucleo-F767ZI equivalent of D6
281 // #define TFT_RST    PF13 // Nucleo-F767ZI equivalent of D7
282
283 // #define TFT_RST    -1  // Set TFT_RST to -1 if the display RESET is connected
    to processor reset
284
    // Use an Arduino pin for initial testing as
    connecting to processor reset
285
    // may not work (pulse too short at power up?)

```



```

286
287  //
    #####
    ###
288  //
289  // Section 3. Define the fonts that are to be used here
290  //
291  //
    #####
    ###
292
293  // Comment out the #defines below with // to stop that font being loaded
294  // The ESP8366 and ESP32 have plenty of memory so commenting out fonts is not
295  // normally necessary. If all fonts are loaded the extra FLASH space required
    is
296  // about 17Kbytes. To save FLASH space only enable the fonts you need!
297
298  #define LOAD_GLCD    // Font 1. Original Adafruit 8 pixel font needs ~1820
    bytes in FLASH
299  #define LOAD_FONT2  // Font 2. Small 16 pixel high font, needs ~3534 bytes in
    FLASH, 96 characters
300  #define LOAD_FONT4  // Font 4. Medium 26 pixel high font, needs ~5848 bytes in
    FLASH, 96 characters
301  #define LOAD_FONT6  // Font 6. Large 48 pixel font, needs ~2666 bytes in
    FLASH, only characters 1234567890:-.apm
302  #define LOAD_FONT7  // Font 7. 7 segment 48 pixel font, needs ~2438 bytes in
    FLASH, only characters 1234567890:-.
303  #define LOAD_FONT8  // Font 8. Large 75 pixel font needs ~3256 bytes in FLASH,
    only characters 1234567890:-.
304  // #define LOAD_FONT8N // Font 8. Alternative to Font 8 above, slightly
    narrower, so 3 digits fit a 160 pixel TFT
305  #define LOAD_GFXFF  // FreeFonts. Include access to the 48 Adafruit_GFX free
    fonts FF1 to FF48 and custom fonts
306
307  // Comment out the #define below to stop the SPIFFS filing system and smooth
    font code being loaded
308  // this will save ~20kbytes of FLASH
309  #define SMOOTH_FONT
310
311  //
    #####
    ###
312  //
313  // Section 4. Other options
314  //
315  //
    #####

```

###

```
316
317 // For RP2040 processor and SPI displays, uncomment the following line to use
    the PIO interface.
318 // #define RP2040_PIO_SPI // Leave commented out to use standard RP2040 SPI
    port interface
319
320 // For RP2040 processor and 8 or 16 bit parallel displays:
321 // The parallel interface write cycle period is derived from a division of the
    CPU clock
322 // speed so scales with the processor clock. This means that the divider ratio
    may need
323 // to be increased when overclocking. I may also need to be adjusted dependant
    on the
324 // display controller type (ILI94341, HX8357C etc). If RP2040_PIO_CLK_DIV is
    not defined
325 // the library will set default values which may not suit your display.
326 // The display controller data sheet will specify the minimum write cycle
    period. The
327 // controllers often work reliably for shorter periods, however if the period
    is too short
328 // the display may not initialise or graphics will become corrupted.
329 // PIO write cycle frequency = (CPU clock / (4 * RP2040_PIO_CLK_DIV))
330 // #define RP2040_PIO_CLK_DIV 1 // 32ns write cycle at 125MHz CPU clock
331 // #define RP2040_PIO_CLK_DIV 2 // 64ns write cycle at 125MHz CPU clock
332 // #define RP2040_PIO_CLK_DIV 3 // 96ns write cycle at 125MHz CPU clock
333
334 // For the RP2040 processor define the SPI port channel used (default 0 if
    undefined)
335 // #define TFT_SPI_PORT 1 // Set to 0 if SPI0 pins are used, or 1 if spi1 pins
    used
336
337 // For the STM32 processor define the SPI port channel used (default 1 if
    undefined)
338 // #define TFT_SPI_PORT 2 // Set to 1 for SPI port 1, or 2 for SPI port 2
339
340 // Define the SPI clock frequency, this affects the graphics rendering speed.
    Too
341 // fast and the TFT driver will not keep up and display corruption appears.
342 // With an ILI9341 display 40MHz works OK, 80MHz sometimes fails
343 // With a ST7735 display more than 27MHz may not work (spurious pixels and
    lines)
344 // With an ILI9163 display 27 MHz works OK.
345
346 // #define SPI_FREQUENCY 10000000
347 // #define SPI_FREQUENCY 50000000
348 // #define SPI_FREQUENCY 100000000
```

```

349 // #define SPI_FREQUENCY 20000000
350 // #define SPI_FREQUENCY 27000000
351 // #define SPI_FREQUENCY 40000000
352 // #define SPI_FREQUENCY 55000000 // STM32 SPI1 only (SPI2 maximum is 27MHz)
353 #define SPI_FREQUENCY 80000000
354
355 // Optional reduced SPI frequency for reading TFT
356 #define SPI_READ_FREQUENCY 20000000
357
358 // The XPT2046 requires a lower SPI clock rate of 2.5MHz so we define that
    here:
359 #define SPI_TOUCH_FREQUENCY 2500000
360
361 // The ESP32 has 2 free SPI ports i.e. VSPI and HSPI, the VSPI is the default.
362 // If the VSPI port is in use and pins are not accessible (e.g. TTGO T-Beam)
363 // then uncomment the following line:
364 // #define USE_HSPI_PORT
365
366 // Comment out the following #define if "SPI Transactions" do not need to be
367 // supported. When commented out the code size will be smaller and sketches
    will
368 // run slightly faster, so leave it commented out unless you need it!
369
370 // Transaction support is needed to work with SD library but not needed with
    TFT_SdFat
371 // Transaction support is required if other SPI devices are connected.
372
373 // Transactions are automatically enabled by the library for an ESP32 (to use
    HAL mutex)
374 // so changing it here has no effect
375
376 // #define SUPPORT_TRANSACTIONS
377

```

2. 将superknob_demo/.pio/libdeps/lolin32_lite/lvgl 目录下的 lv_conf_template.h 文件重命名为 lv_conf.h，并修改其中的部分宏定义使能lvgl的部分功能

lv_conf.h

```

1  /**
2   * @file lv_conf.h
3   * Configuration file for v8.4.0
4   */
5
6  /**
7   * Copy this file as `lv_conf.h`

```

```

8  * 1. simply next to the `lvgl` folder
9  * 2. or any other places and
10 * - define `LV_CONF_INCLUDE_SIMPLE`
11 * - add the path as include path
12 */
13
14 /* clang-format off */
15 #if 1 /*Set it to "1" to enable content*/
16
17 #ifndef LV_CONF_H
18 #define LV_CONF_H
19
20 #include <stdint.h>
21
22 /*=====
23     COLOR SETTINGS
24     =====*/
25
26 /*Color depth: 1 (1 byte per pixel), 8 (RGB332), 16 (RGB565), 32 (ARGB8888)*/
27 #define LV_COLOR_DEPTH 16
28
29 /*Swap the 2 bytes of RGB565 color. Useful if the display has an 8-bit
30 interface (e.g. SPI)*/
31 #define LV_COLOR_16_SWAP 0
32
33 /*Enable features to draw on transparent background.
34 *It's required if opa, and transform_* style properties are used.
35 *Can be also used if the UI is above another layer, e.g. an OSD menu or video
36 player.*/
37 #define LV_COLOR_SCREEN_TRANSP 0
38
39 /* Adjust color mix functions rounding. GPUs might calculate color mix
40 (blending) differently.
41 * 0: round down, 64: round up from x.75, 128: round up from half, 192: round
42 up from x.25, 254: round up */
43 #define LV_COLOR_MIX_ROUND_OFS 0
44
45 /*Images pixels with this color will not be drawn if they are chroma keyed)*/
46 #define LV_COLOR_CHROMA_KEY lv_color_hex(0x00ff00) /*pure green*/
47
48 /*=====
49     MEMORY SETTINGS
50     =====*/
51
52 /*1: use custom malloc/free, 0: use the built-in `lv_mem_alloc()` and
53 `lv_mem_free()` */
54 #define LV_MEM_CUSTOM 0

```

```

50  #if LV_MEM_CUSTOM == 0
51      /*Size of the memory available for `lv_mem_alloc()` in bytes (>= 2kB)*/
52      #define LV_MEM_SIZE (48U * 1024U)          /*[bytes]*/
53
54      /*Set an address for the memory pool instead of allocating it as a normal
array. Can be in external SRAM too.*/
55      #define LV_MEM_ADR 0          /*0: unused*/
56      /*Instead of an address give a memory allocator that will be called to get
a memory pool for LVGL. E.g. my_malloc*/
57      #if LV_MEM_ADR == 0
58          #undef LV_MEM_POOL_INCLUDE
59          #undef LV_MEM_POOL_ALLOC
60      #endif
61
62  #else          /*LV_MEM_CUSTOM*/
63      #define LV_MEM_CUSTOM_INCLUDE <stdlib.h>    /*Header for the dynamic memory
function*/
64      #define LV_MEM_CUSTOM_ALLOC    malloc
65      #define LV_MEM_CUSTOM_FREE    free
66      #define LV_MEM_CUSTOM_REALLOC  realloc
67  #endif        /*LV_MEM_CUSTOM*/
68
69  /*Number of the intermediate memory buffer used during rendering and other
internal processing mechanisms.
70   *You will see an error log message if there wasn't enough buffers. */
71  #define LV_MEM_BUF_MAX_NUM 16
72
73  /*Use the standard `memcpy` and `memset` instead of LVGL's own functions.
(Might or might not be faster).*/
74  #define LV_MEMCPY_MEMSET_STD 0
75
76  /*=====
77      HAL SETTINGS
78  *=====*/
79
80  /*Default display refresh period. LVGL will redraw changed areas with this
period time*/
81  #define LV_DISP_DEF_REFR_PERIOD 30      /*[ms]*/
82
83  /*Input device read period in milliseconds*/
84  #define LV_INDEV_DEF_READ_PERIOD 30     /*[ms]*/
85
86  /*Use a custom tick source that tells the elapsed time in milliseconds.
87   *It removes the need to manually update the tick with `lv_tick_inc()`*/
88  #define LV_TICK_CUSTOM 1
89  #if LV_TICK_CUSTOM

```

```

90     #define LV_TICK_CUSTOM_INCLUDE "Arduino.h"           /*Header for the system
time function*/
91     #define LV_TICK_CUSTOM_SYS_TIME_EXPR (millis())      /*Expression evaluating
to current system time in ms*/
92     /*If using lvgl as ESP32 component*/
93     // #define LV_TICK_CUSTOM_INCLUDE "esp_timer.h"
94     // #define LV_TICK_CUSTOM_SYS_TIME_EXPR ((esp_timer_get_time() / 1000LL))
95 #endif    /*LV_TICK_CUSTOM*/
96
97 /*Default Dot Per Inch. Used to initialize default sizes such as widgets
sized, style paddings.
98  *(Not so important, you can adjust it to modify default sizes and spaces)*/
99 #define LV_DPI_DEF 130    /*[px/inch]*/
100
101 /*=====
102  * FEATURE CONFIGURATION
103  *=====*/
104
105 /*-----
106  * Drawing
107  *-----*/
108
109 /*Enable complex draw engine.
110  *Required to draw shadow, gradient, rounded corners, circles, arc, skew
lines, image transformations or any masks*/
111 #define LV_DRAW_COMPLEX 1
112 #if LV_DRAW_COMPLEX != 0
113
114     /*Allow buffering some shadow calculation.
115     *LV_SHADOW_CACHE_SIZE is the max. shadow size to buffer, where shadow size
is `shadow_width + radius`
116     *Caching has LV_SHADOW_CACHE_SIZE^2 RAM cost*/
117     #define LV_SHADOW_CACHE_SIZE 0
118
119     /* Set number of maximally cached circle data.
120     * The circumference of 1/4 circle are saved for anti-aliasing
121     * radius * 4 bytes are used per circle (the most often used radiuses are
saved)
122     * 0: to disable caching */
123     #define LV_CIRCLE_CACHE_SIZE 4
124 #endif /*LV_DRAW_COMPLEX*/
125
126 /**
127  * "Simple layers" are used when a widget has `style_opa < 255` to buffer the
widget into a layer
128  * and blend it as an image with the given opacity.
129  * Note that `bg_opa`, `text_opa` etc don't require buffering into layer)

```

```

130  * The widget can be buffered in smaller chunks to avoid using large buffers.
131  *
132  * - LV_LAYER_SIMPLE_BUF_SIZE: [bytes] the optimal target buffer size. LVGL
will try to allocate it
133  * - LV_LAYER_SIMPLE_FALLBACK_BUF_SIZE: [bytes] used if
`LV_LAYER_SIMPLE_BUF_SIZE` couldn't be allocated.
134  *
135  * Both buffer sizes are in bytes.
136  * "Transformed layers" (where transform_angle/zoom properties are used) use
larger buffers
137  * and can't be drawn in chunks. So these settings affects only widgets with
opacity.
138  */
139 #define LV_LAYER_SIMPLE_BUF_SIZE          (24 * 1024)
140 #define LV_LAYER_SIMPLE_FALLBACK_BUF_SIZE (3 * 1024)
141
142 /*Default image cache size. Image caching keeps the images opened.
143  *If only the built-in image formats are used there is no real advantage of
caching. (I.e. if no new image decoder is added)
144  *With complex image decoders (e.g. PNG or JPG) caching can save the
continuous open/decode of images.
145  *However the opened images might consume additional RAM.
146  *0: to disable caching*/
147 #define LV_IMG_CACHE_DEF_SIZE 0
148
149 /*Number of stops allowed per gradient. Increase this to allow more stops.
150  *This adds (sizeof(lv_color_t) + 1) bytes per additional stop*/
151 #define LV_GRADIENT_MAX_STOPS 2
152
153 /*Default gradient buffer size.
154  *When LVGL calculates the gradient "maps" it can save them into a cache to
avoid calculating them again.
155  *LV_GRAD_CACHE_DEF_SIZE sets the size of this cache in bytes.
156  *If the cache is too small the map will be allocated only while it's required
for the drawing.
157  *0 mean no caching.*/
158 #define LV_GRAD_CACHE_DEF_SIZE 0
159
160 /*Allow dithering the gradients (to achieve visual smooth color gradients on
limited color depth display)
161  *LV_DITHER_GRADIENT implies allocating one or two more lines of the object's
rendering surface
162  *The increase in memory consumption is (32 bits * object width) plus 24 bits
* object width if using error diffusion */
163 #define LV_DITHER_GRADIENT 0
164 #if LV_DITHER_GRADIENT
165     /*Add support for error diffusion dithering.

```

```

166      *Error diffusion dithering gets a much better visual result, but implies
167      more CPU consumption and memory when drawing.
168      *The increase in memory consumption is (24 bits * object's width)*/
169      #define LV_DITHER_ERROR_DIFFUSION 0
170  #endif
171
172  /*Maximum buffer size to allocate for rotation.
173  *Only used if software rotation is enabled in the display driver.*/
174  #define LV_DISP_ROT_MAX_BUF (10*1024)
175
176  /*-----
177  * GPU
178  *-----*/
179
180  /*Use Arm's 2D acceleration library Arm-2D */
181  #define LV_USE_GPU_ARM2D 0
182
183  /*Use STM32's DMA2D (aka Chrom Art) GPU*/
184  #define LV_USE_GPU_STM32_DMA2D 0
185  #if LV_USE_GPU_STM32_DMA2D
186      /*Must be defined to include path of CMSIS header of target processor
187      e.g. "stm32f7xx.h" or "stm32f4xx.h"*/
188      #define LV_GPU_DMA2D_CMSIS_INCLUDE
189  #endif
190
191  /*Enable RA6M3 G2D GPU*/
192  #define LV_USE_GPU_RA6M3_G2D 0
193  #if LV_USE_GPU_RA6M3_G2D
194      /*include path of target processor
195      e.g. "hal_data.h"*/
196      #define LV_GPU_RA6M3_G2D_INCLUDE "hal_data.h"
197  #endif
198
199  /*Use SWM341's DMA2D GPU*/
200  #define LV_USE_GPU_SWM341_DMA2D 0
201  #if LV_USE_GPU_SWM341_DMA2D
202      #define LV_GPU_SWM341_DMA2D_INCLUDE "SWM341.h"
203  #endif
204
205  /*Use NXP's PXP GPU iMX RTxxx platforms*/
206  #define LV_USE_GPU_NXP_PXP 0
207  #if LV_USE_GPU_NXP_PXP
208      /*1: Add default bare metal and FreeRTOS interrupt handling routines for
209      PXP (lv_gpu_nxp_pxp_osa.c)
210      * and call lv_gpu_nxp_pxp_init() automatically during lv_init(). Note
211      that symbol SDK_OS_FREE_RTOS

```



```

209      *   has to be defined in order to use FreeRTOS OSA, otherwise bare-metal
        implementation is selected.
210      *0: lv_gpu_nxp_pxp_init() has to be called manually before lv_init()
211      */
212      #define LV_USE_GPU_NXP_PXP_AUTO_INIT 0
213 #endif
214
215 /*Use NXP's VG-Lite GPU iMX RTxxx platforms*/
216 #define LV_USE_GPU_NXP_VG_LITE 0
217
218 /*Use SDL renderer API*/
219 #define LV_USE_GPU_SDL 0
220 #if LV_USE_GPU_SDL
221     #define LV_GPU_SDL_INCLUDE_PATH <SDL2/SDL.h>
222     /*Texture cache size, 8MB by default*/
223     #define LV_GPU_SDL_LRU_SIZE (1024 * 1024 * 8)
224     /*Custom blend mode for mask drawing, disable if you need to link with
        older SDL2 lib*/
225     #define LV_GPU_SDL_CUSTOM_BLEND_MODE (SDL_VERSION_ATLEAST(2, 0, 6))
226 #endif
227
228 /*-----
229      * Logging
230      *-----*/
231
232 /*Enable the log module*/
233 #define LV_USE_LOG 0
234 #if LV_USE_LOG
235
236     /*How important log should be added:
237     *LV_LOG_LEVEL_TRACE      A lot of logs to give detailed information
238     *LV_LOG_LEVEL_INFO       Log important events
239     *LV_LOG_LEVEL_WARN       Log if something unwanted happened but didn't
        cause a problem
240     *LV_LOG_LEVEL_ERROR      Only critical issue, when the system may fail
241     *LV_LOG_LEVEL_USER       Only logs added by the user
242     *LV_LOG_LEVEL_NONE       Do not log anything*/
243     #define LV_LOG_LEVEL LV_LOG_LEVEL_WARN
244
245     /*1: Print the log with 'printf';
246     *0: User need to register a callback with `lv_log_register_print_cb()` */
247     #define LV_LOG_PRINTF 0
248
249     /*Enable/disable LV_LOG_TRACE in modules that produces a huge number of
        logs*/
250     #define LV_LOG_TRACE_MEM      1
251     #define LV_LOG_TRACE_TIMER    1

```

```

252     #define LV_LOG_TRACE_INDEV      1
253     #define LV_LOG_TRACE_DISP_REFR  1
254     #define LV_LOG_TRACE_EVENT       1
255     #define LV_LOG_TRACE_OBJ_CREATE  1
256     #define LV_LOG_TRACE_LAYOUT      1
257     #define LV_LOG_TRACE_ANIM        1
258
259 #endif /*LV_USE_LOG*/
260
261 /*-----
262  * Asserts
263  *-----*/
264
265 /*Enable asserts if an operation is failed or an invalid data is found.
266  *If LV_USE_LOG is enabled an error message will be printed on failure*/
267 #define LV_USE_ASSERT_NULL          1 /*Check if the parameter is NULL.
268 (Very fast, recommended)*/
269 #define LV_USE_ASSERT_MALLOC        1 /*Checks is the memory is successfully
270 allocated or no. (Very fast, recommended)*/
271 #define LV_USE_ASSERT_STYLE         0 /*Check if the styles are properly
272 initialized. (Very fast, recommended)*/
273 #define LV_USE_ASSERT_MEM_INTEGRITY 0 /*Check the integrity of `lv_mem`
274 after critical operations. (Slow)*/
275 #define LV_USE_ASSERT_OBJ           0 /*Check the object's type and
276 existence (e.g. not deleted). (Slow)*/
277
278 /*Add a custom handler when assert happens e.g. to restart the MCU*/
279 #define LV_ASSERT_HANDLER_INCLUDE <stdint.h>
280 #define LV_ASSERT_HANDLER while(1); /*Halt by default*/
281
282 /*-----
283  * Others
284  *-----*/
285
286 /*1: Show CPU usage and FPS count*/
287 #define LV_USE_PERF_MONITOR 1
288 #if LV_USE_PERF_MONITOR
289     #define LV_USE_PERF_MONITOR_POS LV_ALIGN_BOTTOM_RIGHT
290 #endif
291
292 /*1: Show the used memory and the memory fragmentation
293  * Requires LV_MEM_CUSTOM = 0*/
294 #define LV_USE_MEM_MONITOR 0
295 #if LV_USE_MEM_MONITOR
296     #define LV_USE_MEM_MONITOR_POS LV_ALIGN_BOTTOM_LEFT
297 #endif
298

```

```

294  /*1: Draw random colored rectangles over the redrawn areas*/
295  #define LV_USE_REFR_DEBUG 0
296
297  /*Change the built in (v)sprintf functions*/
298  #define LV_SPRINTF_CUSTOM 0
299  #if LV_SPRINTF_CUSTOM
300      #define LV_SPRINTF_INCLUDE <stdio.h>
301      #define lv_sprintf  sprintf
302      #define lv_vsprintf vsprintf
303  #else    /*LV_SPRINTF_CUSTOM*/
304      #define LV_SPRINTF_USE_FLOAT 0
305  #endif  /*LV_SPRINTF_CUSTOM*/
306
307  #define LV_USE_USER_DATA 1
308
309  /*Garbage Collector settings
310   *Used if lvgl is bound to higher level language and the memory is managed by
311   *that language*/
312  #define LV_ENABLE_GC 0
313  #if LV_ENABLE_GC != 0
314      #define LV_GC_INCLUDE "gc.h"                /*Include Garbage
315      Collector related things*/
316  #endif /*LV_ENABLE_GC*/
317
318  /*=====
319  *  COMPILER SETTINGS
320  *=====*/
321
322  /*For big endian systems set to 1*/
323  #define LV_BIG_ENDIAN_SYSTEM 0
324
325  /*Define a custom attribute to `lv_tick_inc` function*/
326  #define LV_ATTRIBUTE_TICK_INC
327
328  /*Define a custom attribute to `lv_timer_handler` function*/
329  #define LV_ATTRIBUTE_TIMER_HANDLER
330
331  /*Define a custom attribute to `lv_disp_flush_ready` function*/
332  #define LV_ATTRIBUTE_FLUSH_READY
333
334  /*Required alignment size for buffers*/
335  #define LV_ATTRIBUTE_MEM_ALIGN_SIZE 1
336
337  /*Will be added where memories needs to be aligned (with -Os data might not be
338  aligned to boundary by default).
339   * E.g. __attribute__((aligned(4)))*/
340  #define LV_ATTRIBUTE_MEM_ALIGN

```

```
338
339  /*Attribute to mark large constant arrays for example font's bitmaps*/
340  #define LV_ATTRIBUTE_LARGE_CONST
341
342  /*Compiler prefix for a big array declaration in RAM*/
343  #define LV_ATTRIBUTE_LARGE_RAM_ARRAY
344
345  /*Place performance critical functions into a faster memory (e.g RAM)*/
346  #define LV_ATTRIBUTE_FAST_MEM
347
348  /*Prefix variables that are used in GPU accelerated operations, often these
349  need to be placed in RAM sections that are DMA accessible*/
349  #define LV_ATTRIBUTE_DMA
350
351  /*Export integer constant to binding. This macro is used with constants in the
352  form of LV_<CONST> that
353    *should also appear on LVGL binding API such as Micropython.*/
353  #define LV_EXPORT_CONST_INT(int_value) struct _silence_gcc_warning /*The
354  default value just prevents GCC warning*/
355
355  /*Extend the default -32k..32k coordinate range to -4M..4M by using int32_t
356  for coordinates instead of int16_t*/
356  #define LV_USE_LARGE_COORD 0
357
358  /*=====
359    *   FONT USAGE
360    *=====*/
361
362  /*Montserrat fonts with ASCII range and some symbols using bpp = 4
363    *https://fonts.google.com/specimen/Montserrat*/
364  #define LV_FONT_MONTSERRAT_8 0
365  #define LV_FONT_MONTSERRAT_10 0
366  #define LV_FONT_MONTSERRAT_12 0
367  #define LV_FONT_MONTSERRAT_14 1
368  #define LV_FONT_MONTSERRAT_16 0
369  #define LV_FONT_MONTSERRAT_18 0
370  #define LV_FONT_MONTSERRAT_20 0
371  #define LV_FONT_MONTSERRAT_22 0
372  #define LV_FONT_MONTSERRAT_24 0
373  #define LV_FONT_MONTSERRAT_26 0
374  #define LV_FONT_MONTSERRAT_28 0
375  #define LV_FONT_MONTSERRAT_30 0
376  #define LV_FONT_MONTSERRAT_32 0
377  #define LV_FONT_MONTSERRAT_34 0
378  #define LV_FONT_MONTSERRAT_36 0
379  #define LV_FONT_MONTSERRAT_38 0
380  #define LV_FONT_MONTSERRAT_40 0
```

```

381 #define LV_FONT_MONTERRAT_42 0
382 #define LV_FONT_MONTERRAT_44 0
383 #define LV_FONT_MONTERRAT_46 0
384 #define LV_FONT_MONTERRAT_48 0
385
386 /*Demonstrate special features*/
387 #define LV_FONT_MONTERRAT_12_SUBPX 0
388 #define LV_FONT_MONTERRAT_28_COMPRESSED 0 /*bpp = 3*/
389 #define LV_FONT_DEJAVU_16_PERSIAN_HEBREW 0 /*Hebrew, Arabic, Persian letters
and all their forms*/
390 #define LV_FONT_SIMSUN_16_CJK 0 /*1000 most common CJK radicals*/
391
392 /*Pixel perfect monospace fonts*/
393 #define LV_FONT_UNSCII_8 0
394 #define LV_FONT_UNSCII_16 0
395
396 /*Optionally declare custom fonts here.
397  *You can use these fonts as default font too and they will be available
globally.
398  *E.g. #define LV_FONT_CUSTOM_DECLARE LV_FONT_DECLARE(my_font_1)
LV_FONT_DECLARE(my_font_2)*/
399 #define LV_FONT_CUSTOM_DECLARE
400
401 /*Always set a default font*/
402 #define LV_FONT_DEFAULT &lv_font_montserrat_14
403
404 /*Enable handling large font and/or fonts with a lot of characters.
405  *The limit depends on the font size, font face and bpp.
406  *Compiler error will be triggered if a font needs it.*/
407 #define LV_FONT_FMT_TXT_LARGE 0
408
409 /*Enables/disables support for compressed fonts.*/
410 #define LV_USE_FONT_COMPRESSED 0
411
412 /*Enable subpixel rendering*/
413 #define LV_USE_FONT_SUBPX 0
414 #if LV_USE_FONT_SUBPX
415     /*Set the pixel order of the display. Physical order of RGB channels.
Doesn't matter with "normal" fonts.*/
416     #define LV_FONT_SUBPX_BGR 0 /*0: RGB; 1:BGR order*/
417 #endif
418
419 /*Enable drawing placeholders when glyph dsc is not found*/
420 #define LV_USE_FONT_PLACEHOLDER 1
421
422 /*=====
423  * TEXT SETTINGS

```

```

424  *=====*/
425
426  /**
427   * Select a character encoding for strings.
428   * Your IDE or editor should have the same character encoding
429   * - LV_TXT_ENC_UTF8
430   * - LV_TXT_ENC_ASCII
431   */
432  #define LV_TXT_ENC LV_TXT_ENC_UTF8
433
434  /*Can break (wrap) texts on these chars*/
435  #define LV_TXT_BREAK_CHARS " ,.;:-_"
436
437  /*If a word is at least this long, will break wherever "prettiest"
438   *To disable, set to a value <= 0*/
439  #define LV_TXT_LINE_BREAK_LONG_LEN 0
440
441  /*Minimum number of characters in a long word to put on a line before a break.
442   *Depends on LV_TXT_LINE_BREAK_LONG_LEN.*/
443  #define LV_TXT_LINE_BREAK_LONG_PRE_MIN_LEN 3
444
445  /*Minimum number of characters in a long word to put on a line after a break.
446   *Depends on LV_TXT_LINE_BREAK_LONG_LEN.*/
447  #define LV_TXT_LINE_BREAK_LONG_POST_MIN_LEN 3
448
449  /*The control character to use for signalling text recoloring.*/
450  #define LV_TXT_COLOR_CMD "#"
451
452  /*Support bidirectional texts. Allows mixing Left-to-Right and Right-to-Left
453   texts.
454   *The direction will be processed according to the Unicode Bidirectional
455   Algorithm:
456   *https://www.w3.org/International/articles/inline-bidi-markup/uba-basics*/
457  #define LV_USE_BIDI 0
458  #if LV_USE_BIDI
459      /*Set the default direction. Supported values:
460       *`LV_BASE_DIR_LTR` Left-to-Right
461       *`LV_BASE_DIR_RTL` Right-to-Left
462       *`LV_BASE_DIR_AUTO` detect texts base direction*/
463      #define LV_BIDI_BASE_DIR_DEF LV_BASE_DIR_AUTO
464  #endif
465
466  /*Enable Arabic/Persian processing
467   *In these languages characters should be replaced with an other form based on
468   their position in the text*/
469  #define LV_USE_ARABIC_PERSIAN_CHARS 0
470

```

```

468  /*=====
469  *  WIDGET USAGE
470  *=====*/
471
472  /*Documentation of the widgets:
473  https://docs.lvgl.io/latest/en/html/widgets/index.html*/
474
474  #define LV_USE_ARC          1
475
476  #define LV_USE_BAR          1
477
478  #define LV_USE_BTN          1
479
480  #define LV_USE_BTNMATRIX   1
481
482  #define LV_USE_CANVAS      1
483
484  #define LV_USE_CHECKBOX    1
485
486  #define LV_USE_DROPDOWN    1  /*Requires: lv_label*/
487
488  #define LV_USE_IMG          1  /*Requires: lv_label*/
489
490  #define LV_USE_LABEL        1
491  #if LV_USE_LABEL
492      #define LV_LABEL_TEXT_SELECTION 1 /*Enable selecting text of the label*/
493      #define LV_LABEL_LONG_TXT_HINT 1 /*Store some extra info in labels to
494                                     speed up drawing of very long texts*/
495  #endif
496
497  #define LV_USE_LINE          1
498
499  #define LV_USE_ROLLER        1  /*Requires: lv_label*/
500  #if LV_USE_ROLLER
501      #define LV_ROLLER_INF_PAGES 7 /*Number of extra "pages" when the roller is
502                                     infinite*/
503  #endif
504
505  #define LV_USE_SLIDER        1  /*Requires: lv_bar*/
506
507  #define LV_USE_SWITCH        1
508
509  #define LV_USE_TEXTAREA      1  /*Requires: lv_label*/
510  #if LV_USE_TEXTAREA != 0
511      #define LV_TEXTAREA_DEF_PWD_SHOW_TIME 1500 /*ms*/
512  #endif
513

```

```

512 #define LV_USE_TABLE      1
513
514 /*=====
515  * EXTRA COMPONENTS
516  *=====*/
517
518 /*-----
519  * Widgets
520  *-----*/
521 #define LV_USE_ANIMIMG     1
522
523 #define LV_USE_CALENDAR    1
524 #if LV_USE_CALENDAR
525     #define LV_CALENDAR_WEEK_STARTS_MONDAY 0
526     #if LV_CALENDAR_WEEK_STARTS_MONDAY
527         #define LV_CALENDAR_DEFAULT_DAY_NAMES {"Mo", "Tu", "We", "Th", "Fr",
528 "Sa", "Su"}
529     #else
530         #define LV_CALENDAR_DEFAULT_DAY_NAMES {"Su", "Mo", "Tu", "We", "Th",
531 "Fr", "Sa"}
532     #endif
533     #define LV_CALENDAR_DEFAULT_MONTH_NAMES {"January", "February", "March",
534 "April", "May", "June", "July", "August", "September", "October", "November",
535 "December"}
536     #define LV_USE_CALENDAR_HEADER_ARROW 1
537     #define LV_USE_CALENDAR_HEADER_DROPDOWN 1
538 #endif /*LV_USE_CALENDAR*/
539
540 #define LV_USE_CHART      1
541
542 #define LV_USE_COLORWHEEL 1
543
544 #define LV_USE_IMGBTN     1
545
546 #define LV_USE_KEYBOARD   1
547
548 #define LV_USE_LED        1
549
550 #define LV_USE_LIST       1
551
552 #define LV_USE_MENU       1
553
554 #define LV_USE_METER      1
555
556 #define LV_USE_MSGBOX     1
557
558 #define LV_USE_SLIDER     1
559
560 #define LV_USE_SPINBOX    1
561
562 #define LV_USE_SPINNER    1
563
564 #define LV_USE_SWITCH     1
565
566 #define LV_USE_TABVIEW    1
567
568 #define LV_USE_TILEVIEW   1
569
570 #define LV_USE_WIN        1
571
572 #define LV_USE_ZOOMER     1
573
574 #define LV_USE_ZOOMER     1
575
576 #define LV_USE_ZOOMER     1
577
578 #define LV_USE_ZOOMER     1
579
580 #define LV_USE_ZOOMER     1
581
582 #define LV_USE_ZOOMER     1
583
584 #define LV_USE_ZOOMER     1
585
586 #define LV_USE_ZOOMER     1
587
588 #define LV_USE_ZOOMER     1
589
590 #define LV_USE_ZOOMER     1
591
592 #define LV_USE_ZOOMER     1
593
594 #define LV_USE_ZOOMER     1
595
596 #define LV_USE_ZOOMER     1
597
598 #define LV_USE_ZOOMER     1
599
600 #define LV_USE_ZOOMER     1
601
602 #define LV_USE_ZOOMER     1
603
604 #define LV_USE_ZOOMER     1
605
606 #define LV_USE_ZOOMER     1
607
608 #define LV_USE_ZOOMER     1
609
610 #define LV_USE_ZOOMER     1
611
612 #define LV_USE_ZOOMER     1
613
614 #define LV_USE_ZOOMER     1
615
616 #define LV_USE_ZOOMER     1
617
618 #define LV_USE_ZOOMER     1
619
620 #define LV_USE_ZOOMER     1
621
622 #define LV_USE_ZOOMER     1
623
624 #define LV_USE_ZOOMER     1
625
626 #define LV_USE_ZOOMER     1
627
628 #define LV_USE_ZOOMER     1
629
630 #define LV_USE_ZOOMER     1
631
632 #define LV_USE_ZOOMER     1
633
634 #define LV_USE_ZOOMER     1
635
636 #define LV_USE_ZOOMER     1
637
638 #define LV_USE_ZOOMER     1
639
640 #define LV_USE_ZOOMER     1
641
642 #define LV_USE_ZOOMER     1
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644 #define LV_USE_ZOOMER     1
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646 #define LV_USE_ZOOMER     1
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648 #define LV_USE_ZOOMER     1
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650 #define LV_USE_ZOOMER     1
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652 #define LV_USE_ZOOMER     1
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654 #define LV_USE_ZOOMER     1
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656 #define LV_USE_ZOOMER     1
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658 #define LV_USE_ZOOMER     1
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660 #define LV_USE_ZOOMER     1
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662 #define LV_USE_ZOOMER     1
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664 #define LV_USE_ZOOMER     1
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666 #define LV_USE_ZOOMER     1
667
668 #define LV_USE_ZOOMER     1
669
670 #define LV_USE_ZOOMER     1
671
672 #define LV_USE_ZOOMER     1
673
674 #define LV_USE_ZOOMER     1
675
676 #define LV_USE_ZOOMER     1
677
678 #define LV_USE_ZOOMER     1
679
680 #define LV_USE_ZOOMER     1
681
682 #define LV_USE_ZOOMER     1
683
684 #define LV_USE_ZOOMER     1
685
686 #define LV_USE_ZOOMER     1
687
688 #define LV_USE_ZOOMER     1
689
690 #define LV_USE_ZOOMER     1
691
692 #define LV_USE_ZOOMER     1
693
694 #define LV_USE_ZOOMER     1
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555 #define LV_USE_SPAN      1
556 #if LV_USE_SPAN
557     /*A line text can contain maximum num of span descriptor */
558     #define LV_SPAN_SNIPPET_STACK_SIZE 64
559 #endif
560
561 #define LV_USE_SPINBOX    1
562
563 #define LV_USE_SPINNER    1
564
565 #define LV_USE_TABVIEW    1
566
567 #define LV_USE_TILEVIEW   1
568
569 #define LV_USE_WIN        1
570
571 /*-----
572  * Themes
573  *-----*/
574
575 /*A simple, impressive and very complete theme*/
576 #define LV_USE_THEME_DEFAULT 1
577 #if LV_USE_THEME_DEFAULT
578
579     /*0: Light mode; 1: Dark mode*/
580     #define LV_THEME_DEFAULT_DARK 0
581
582     /*1: Enable grow on press*/
583     #define LV_THEME_DEFAULT_GROW 1
584
585     /*Default transition time in [ms]*/
586     #define LV_THEME_DEFAULT_TRANSITION_TIME 80
587 #endif /*LV_USE_THEME_DEFAULT*/
588
589 /*A very simple theme that is a good starting point for a custom theme*/
590 #define LV_USE_THEME_BASIC 1
591
592 /*A theme designed for monochrome displays*/
593 #define LV_USE_THEME_MONO 1
594
595 /*-----
596  * Layouts
597  *-----*/
598
599 /*A layout similar to Flexbox in CSS.*/
600 #define LV_USE_FLEX 1
601

```

```

602  /*A layout similar to Grid in CSS.*/
603  #define LV_USE_GRID 1
604
605  /*-----
606   * 3rd party libraries
607   *-----*/
608
609  /*File system interfaces for common APIs */
610
611  /*API for fopen, fread, etc*/
612  #define LV_USE_FS_STDIO 0
613  #if LV_USE_FS_STDIO
614      #define LV_FS_STDIO_LETTER '\0' /*Set an upper cased letter on which
the drive will accessible (e.g. 'A')*/
615      #define LV_FS_STDIO_PATH "" /*Set the working directory.
File/directory paths will be appended to it.*/
616      #define LV_FS_STDIO_CACHE_SIZE 0 /*>0 to cache this number of bytes in
lv_fs_read()*/
617  #endif
618
619  /*API for open, read, etc*/
620  #define LV_USE_FS_POSIX 0
621  #if LV_USE_FS_POSIX
622      #define LV_FS_POSIX_LETTER '\0' /*Set an upper cased letter on which
the drive will accessible (e.g. 'A')*/
623      #define LV_FS_POSIX_PATH "" /*Set the working directory.
File/directory paths will be appended to it.*/
624      #define LV_FS_POSIX_CACHE_SIZE 0 /*>0 to cache this number of bytes in
lv_fs_read()*/
625  #endif
626
627  /*API for CreateFile, ReadFile, etc*/
628  #define LV_USE_FS_WIN32 0
629  #if LV_USE_FS_WIN32
630      #define LV_FS_WIN32_LETTER '\0' /*Set an upper cased letter on which
the drive will accessible (e.g. 'A')*/
631      #define LV_FS_WIN32_PATH "" /*Set the working directory.
File/directory paths will be appended to it.*/
632      #define LV_FS_WIN32_CACHE_SIZE 0 /*>0 to cache this number of bytes in
lv_fs_read()*/
633  #endif
634
635  /*API for FATFS (needs to be added separately). Uses f_open, f_read, etc*/
636  #define LV_USE_FS_FATFS 0
637  #if LV_USE_FS_FATFS
638      #define LV_FS_FATFS_LETTER '\0' /*Set an upper cased letter on which
the drive will accessible (e.g. 'A')*/

```

```

639     #define LV_FS_FATFS_CACHE_SIZE 0    /*>0 to cache this number of bytes in
lv_fs_read()*/
640 #endif
641
642 /*API for LittleFS (library needs to be added separately). Uses lfs_file_open,
lfs_file_read, etc*/
643 #define LV_USE_FS_LITTLEFS 0
644 #if LV_USE_FS_LITTLEFS
645     #define LV_FS_LITTLEFS_LETTER '\0'    /*Set an upper cased letter on
which the drive will be accessible (e.g. 'A')*/
646     #define LV_FS_LITTLEFS_CACHE_SIZE 0    /*>0 to cache this number of bytes
in lv_fs_read()*/
647 #endif
648
649 /*PNG decoder library*/
650 #define LV_USE_PNG 0
651
652 /*BMP decoder library*/
653 #define LV_USE_BMP 0
654
655 /*JPG + split JPG decoder library.
 * Split JPG is a custom format optimized for embedded systems. */
656 #define LV_USE_SJPG 0
657
658 /*GIF decoder library*/
659 #define LV_USE_GIF 0
660
661 /*QR code library*/
662 #define LV_USE_QRCODE 0
663
664 /*FreeType library*/
665 #define LV_USE_FREETYPE 0
666 #if LV_USE_FREETYPE
667     /*Memory used by FreeType to cache characters [bytes] (-1: no caching)*/
668     #define LV_FREETYPE_CACHE_SIZE (16 * 1024)
669     #if LV_FREETYPE_CACHE_SIZE >= 0
670         /* 1: bitmap cache use the sbit cache, 0: bitmap cache use the image
cache. */
671         /* sbit cache: it is much more memory efficient for small bitmaps (font
size < 256) */
672         /* if font size >= 256, must be configured as image cache */
673         #define LV_FREETYPE_SBIT_CACHE 0
674         /* Maximum number of opened FT_Face/FT_Size objects managed by this
cache instance. */
675         /* (0: use system defaults) */
676         #define LV_FREETYPE_CACHE_FT_FACES 0
677         #define LV_FREETYPE_CACHE_FT_SIZES 0

```

```
679     #endif
680 #endif
681
682 /*Tiny TTF library*/
683 #define LV_USE_TINY_TTF 0
684 #if LV_USE_TINY_TTF
685     /*Load TTF data from files*/
686     #define LV_TINY_TTF_FILE_SUPPORT 0
687 #endif
688
689 /*Rlottie library*/
690 #define LV_USE_RLOTTIE 0
691
692 /*FFmpeg library for image decoding and playing videos
693  *Supports all major image formats so do not enable other image decoder with
694  it*/
695 #define LV_USE_FFMPEG 0
696 #if LV_USE_FFMPEG
697     /*Dump input information to stderr*/
698     #define LV_FFMPEG_DUMP_FORMAT 0
699 #endif
700
701 /*-----
702  * Others
703  *-----*/
704
705 /*1: Enable API to take snapshot for object*/
706 #define LV_USE_SNAPSHOT 0
707
708 /*1: Enable Monkey test*/
709 #define LV_USE_MONKEY 0
710
711 /*1: Enable grid navigation*/
712 #define LV_USE_GRIDNAV 0
713
714 /*1: Enable lv_obj fragment*/
715 #define LV_USE_FRAGMENT 0
716
717 /*1: Support using images as font in label or span widgets */
718 #define LV_USE_IMGFONT 0
719
720 /*1: Enable a published subscriber based messaging system */
721 #define LV_USE_MSG 0
722
723 /*1: Enable Pinyin input method*/
724 /*Requires: lv_keyboard*/
725 #define LV_USE_IME_PINYIN 0
```

```

725 #if LV_USE_IME_PINYIN
726     /*1: Use default thesaurus*/
727     /*If you do not use the default thesaurus, be sure to use `lv_ime_pinyin`
after setting the thesaurus*/
728     #define LV_IME_PINYIN_USE_DEFAULT_DICT 1
729     /*Set the maximum number of candidate panels that can be displayed*/
730     /*This needs to be adjusted according to the size of the screen*/
731     #define LV_IME_PINYIN_CAND_TEXT_NUM 6
732
733     /*Use 9 key input(k9)*/
734     #define LV_IME_PINYIN_USE_K9_MODE      1
735     #if LV_IME_PINYIN_USE_K9_MODE == 1
736         #define LV_IME_PINYIN_K9_CAND_TEXT_NUM 3
737     #endif // LV_IME_PINYIN_USE_K9_MODE
738 #endif
739
740 /*=====
741  * EXAMPLES
742  *=====*/
743
744 /*Enable the examples to be built with the library*/
745 #define LV_BUILD_EXAMPLES 1
746
747 /*=====
748  * DEMO USAGE
749  *=====*/
750
751 /*Show some widget. It might be required to increase `LV_MEM_SIZE` */
752 #define LV_USE_DEMO_WIDGETS 0
753 #if LV_USE_DEMO_WIDGETS
754     #define LV_DEMO_WIDGETS_SLIDESHOW 0
755 #endif
756
757 /*Demonstrate the usage of encoder and keyboard*/
758 #define LV_USE_DEMO_KEYPAD_AND_ENCODER 0
759
760 /*Benchmark your system*/
761 #define LV_USE_DEMO_BENCHMARK 1
762 #if LV_USE_DEMO_BENCHMARK
763     /*Use RGB565A8 images with 16 bit color depth instead of ARGB8565*/
764     #define LV_DEMO_BENCHMARK_RGB565A8 0
765 #endif
766
767 /*Stress test for LVGL*/
768 #define LV_USE_DEMO_STRESS 0
769
770 /*Music player demo*/

```

```
771 #define LV_USE_DEMO_MUSIC 0
772 #if LV_USE_DEMO_MUSIC
773     #define LV_DEMO_MUSIC_SQUARE 0
774     #define LV_DEMO_MUSIC_LANDSCAPE 0
775     #define LV_DEMO_MUSIC_ROUND 0
776     #define LV_DEMO_MUSIC_LARGE 0
777     #define LV_DEMO_MUSIC_AUTO_PLAY 0
778 #endif
779
780 /*--END OF LV_CONF_H--*/
781
782 #endif /*LV_CONF_H*/
783
784 #endif /*End of "Content enable"*/
785
```